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# Upnormal behavior detection

CSE/AIE Graduation Project Presentation

**Supervised by : Dr Essam Abdellatef**

15/6/2026



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# SecureSight — Real-Time Intelligent Surveillance System for Video Anomaly Detection



## 1. Massive CCTV footage is under-analyzed

Vast amounts of recorded video data remain unreviewed in real time, leading to potential oversights.



## 2. Human monitoring suffers from attention fatigue

Operators cannot maintain high vigilance for extended periods, increasing the risk of missed incidents.



## 3. Delayed incident detection and response

Critical anomalies are often detected too late, slowing down necessary interventions and reaction times.



## 4. Lack of smart role-specific emergency alert routing

Many systems fail to automatically direct alerts to the appropriate authorities based on the type of incident.

**The challenge is not only detecting anomalies, but detecting them fast, reliably, and sending the alert to the right responder.**

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# From passive recording to proactive real-time emergency response.

- Faster detection of dangerous or abnormal activities
- Reduced dependence on continuous human monitoring
- Automatic routing of alerts
- Better operational reliability through offline alert buffering
- Support safer public spaces



# Research Background & The SecureSight Gap

### 1. Reconstruction-Based Methods

- Autoencoders detect anomalies using reconstruction error.

⚠ Difficult to reconstruct diverse, complex anomalies.

### 2. Weakly Supervised MIL Methods

- UCF-Crime introduced video-level anomaly learning.

⚠ Coarse video-level labels limit precise spatial localization.

### 3. Transformer-Based Video Models

- Capture long-range temporal dependencies.

⚠ High computational cost and data requirements.

### 4. Two-Stream Networks

- Combine spatial appearance and temporal motion.

⚠ Research Gap: Efficient real-time scaling and feature fusion for multi-scale context-awareness.

Current literature lacks an efficient approach to integrate multi-scale spatial and temporal context awareness for precise, real-time surveillance anomaly detection, a critical requirement for complex scenes.

# Recent Published papers limitations

| Paper | Data Issues | Bias | Evaluation | Generalization | Model Issues | Temporal Issues | Training Issues | System / Deployment | Efficiency | Other |
|---|---|---|---|---|---|---|---|---|---|---|
| Shoplifting (YOLOv5) | Dataset, Representativeness, Imbalance, Truncation | Bias | Metrics | Generalization | Features, Robustness, Tracking | — | — | Deployment | — | — |
| Survey (Suspicious Behavior) | — | Bias | Metrics, Evaluation | — | Modality | — | — | Deployment | — | Selection, Coverage, Rigor, Definition, Ethics, Clarity |
| Spatiotemporal Autoencoder | Small Datasets, Low Diversity | Reconstruction Bias | — | Poor Generalization | — | — | Manual Thresholds | — | — | Fixed Cameras, No Ablation |
| Wu et al. (2024) | — | Benchmark Bias | Weak Analysis | — | — | — | — | — | — | Inconsistent Comparisons, No Meta-analysis, Limited Coverage |
| Song (2022) | — | Selection Bias | Weak Evaluation, No Metrics | — | — | — | — | — | — | Shallow Review, No Failure Analysis |
| Real-world AD | Limited Data, Class Imbalance | — | — | Weak Generalization | — | — | Sparse Training | — | — | No Fairness |
| RTFM | — | — | — | No Generalization | Backbone Dependency | — | Label Noise | — | — | Fragile Assumptions, Fixed Segmentation |
| Object-Augmented VAD | — | COCO Bias | — | Weak Generalization | — | Temporal Misalignment | — | — | No FPS | Top-K Limitation |
| Future Frame Prediction | — | — | PSNR Weakness | — | — | Short Horizon | — | — | — | Low Resolution, No Uncertainty, Flow Assumptions |
| ViT | Data Hungry | — | — | — | High Complexity | — | — | No Deployment | Scaling Issues | Poor Small Data |
| Our Paper | — | — | Weak Evaluation | — | — | — | — | Prototype System | Latency Gaps | Unfair Comparison, Non-standard Split |

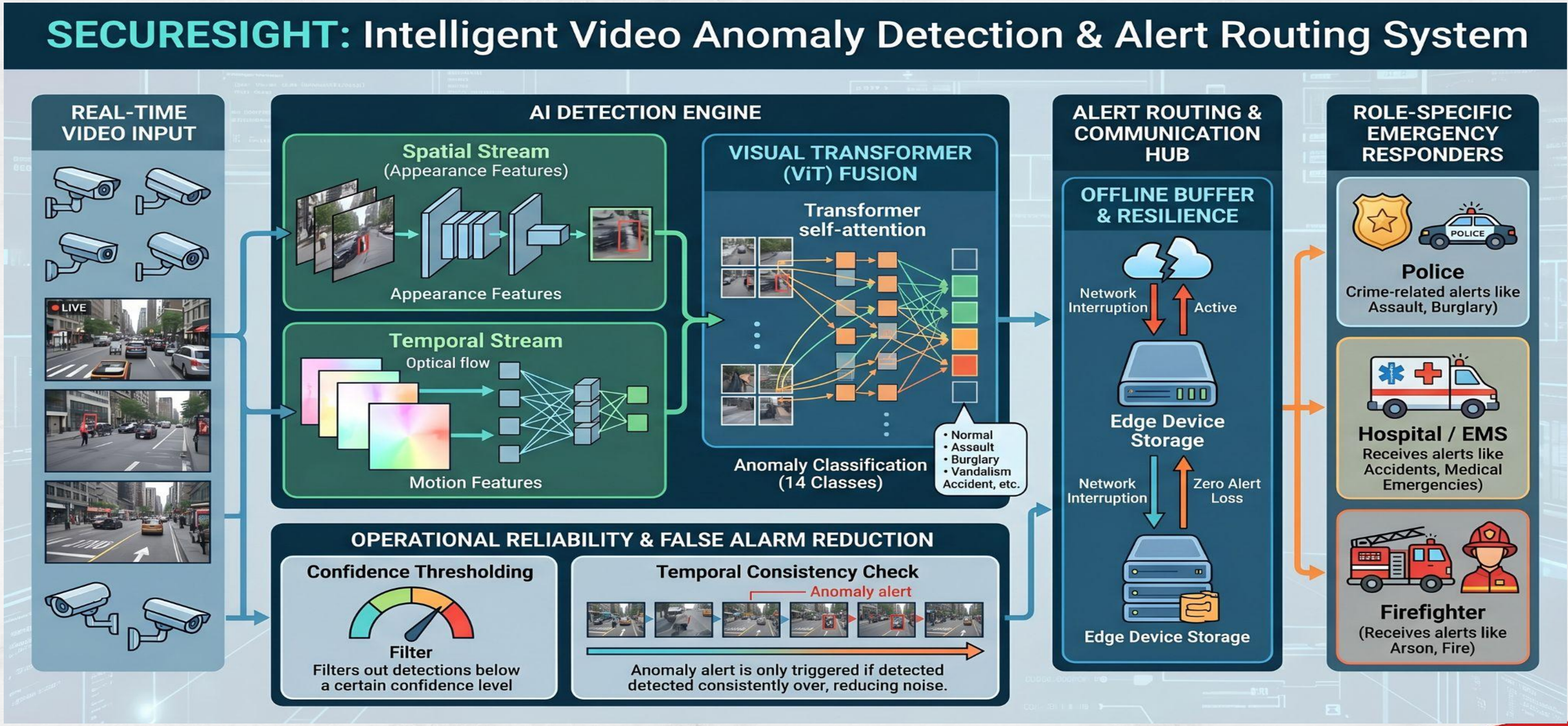
# SecureSight — Real-Time Intelligent Surveillance System for Video Anomaly Detection

Based on a literature review and limitations of recent published papers, this slide summarizes the key research gaps that motivated our proposed system.

- Weak Real-Time Deployment
- Limited Alert Routing
- Incomplete Spatial-Temporal Understanding
- Weak Evaluation and Generalization
- Reliability Gap

**Therefore, SecureSight was designed as an end-to-end system** combining spatial-temporal anomaly detection, ViT-based temporal reasoning, MQTT alert routing, role-specific dashboards, and offline alert buffering.

# Proposed Solution



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# Transforming surveillance from passive recording into real-time intelligent understanding and emergency alerting.

## TRADITIONAL SURVEILLANCE (Passive)

- Video Recording Only
- Human-Dependent Monitoring
- Delayed Response
- Missed Incidents

## SECURESIGHT (Intelligent & Real-Time)

- Real-Time Video Understanding
- Automatic Anomaly Detection
- Instant Alert Generation
- Smart Routing



# Requirements

## Functional Requirements

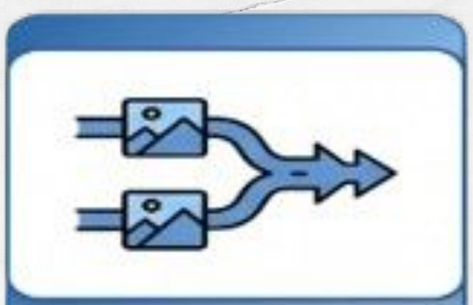
- Accept CCTV input
- Extract clips
- Preprocess video
- Classify activity
- Generate alerts
- Route alerts to police/EMS
- Buffer alerts

## Non-Functional Requirements

- Low-latency inference
- Reliable alert delivery
- Scalable architecture
- Role-specific dashboards
- Maintainable design
- Privacy-aware handling

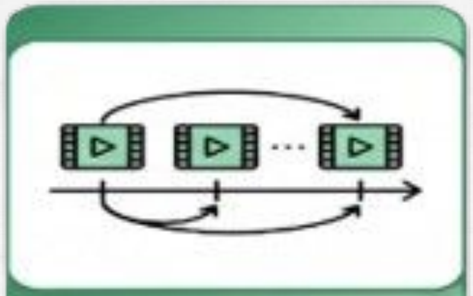
**Requirements combine AI accuracy with real-world system reliability.**

# Why our solution is strong



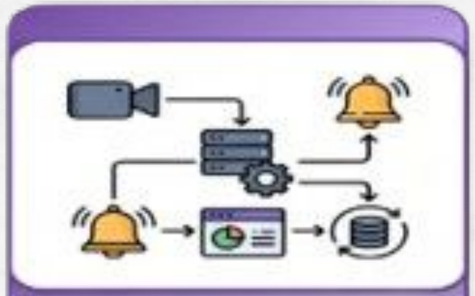
**Two-Stream Understanding**

Combines spatial appearance and temporal motion.



**ViT Temporal Reasoning**

Captures long-range dependencies across video clips.



**Real-Time Deployment Design**

Flask inference, MQTT alert routing, dashboards, and offline buffer.

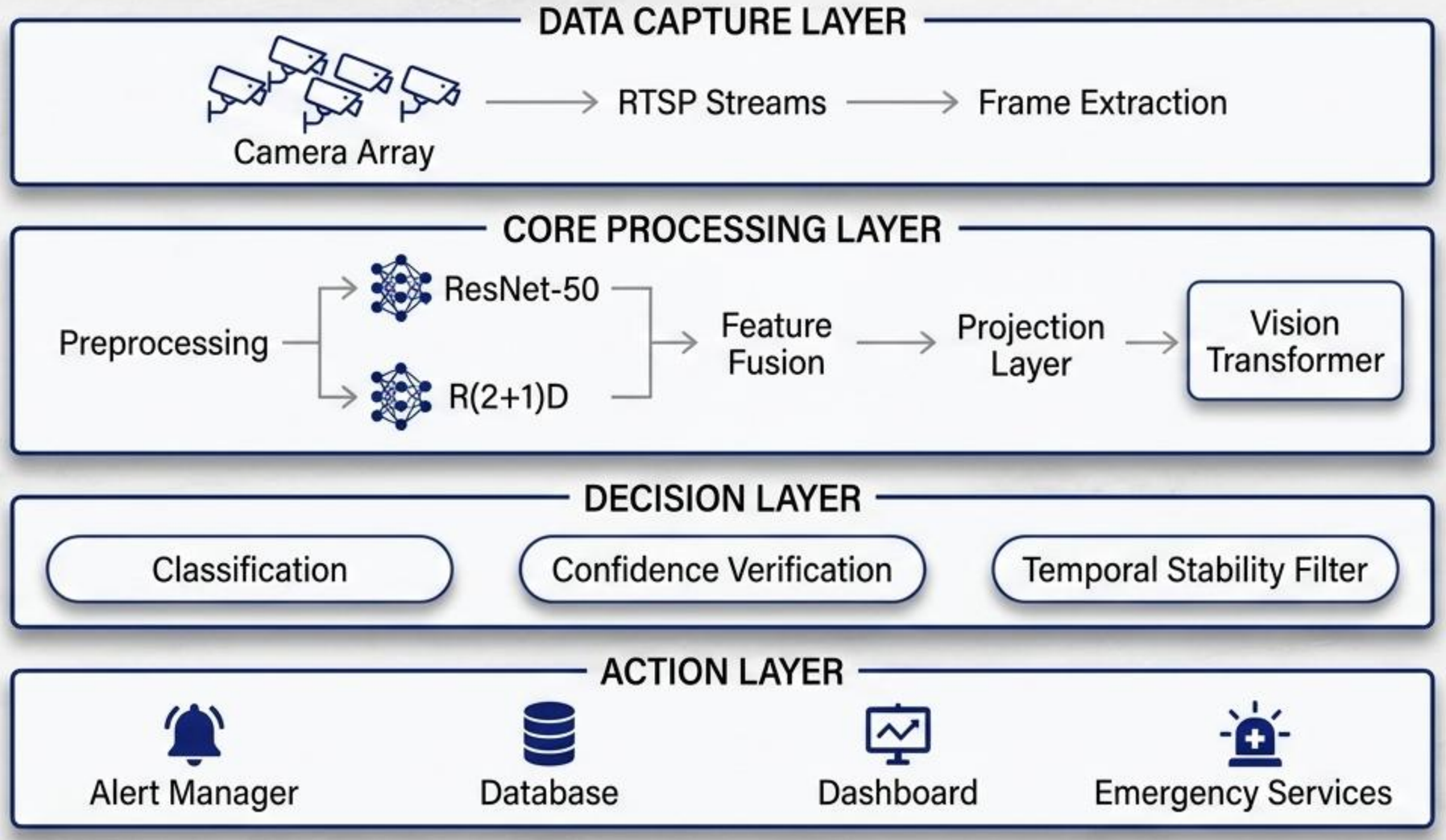


**Validated Performance**

74.3% top-1 accuracy and 0.89 mean AUC on UCF-Crime.

**The project is not only a model; it is an end-to-end intelligent surveillance prototype.**

# System Architecture Layers



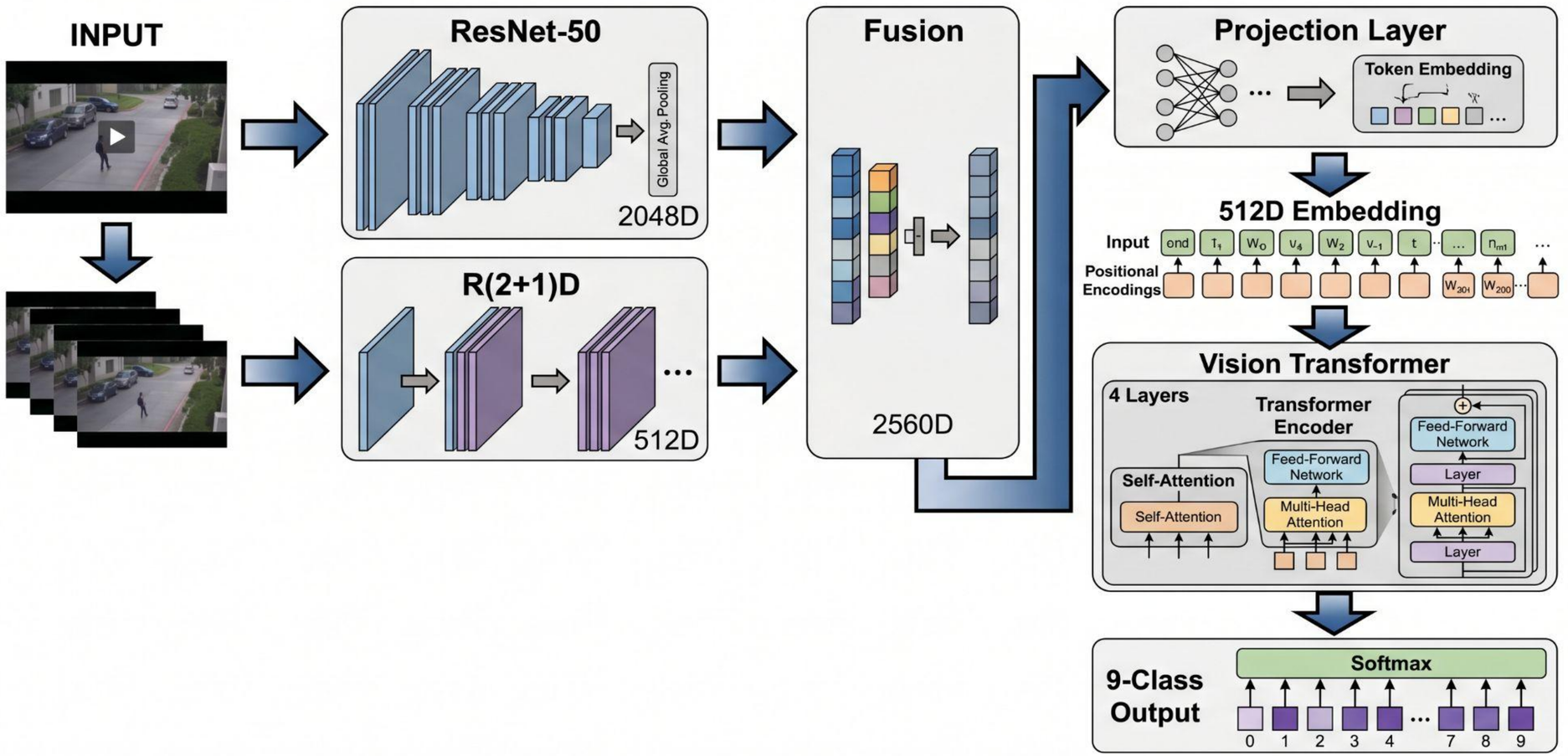
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# SYSTEM WORKFLOW

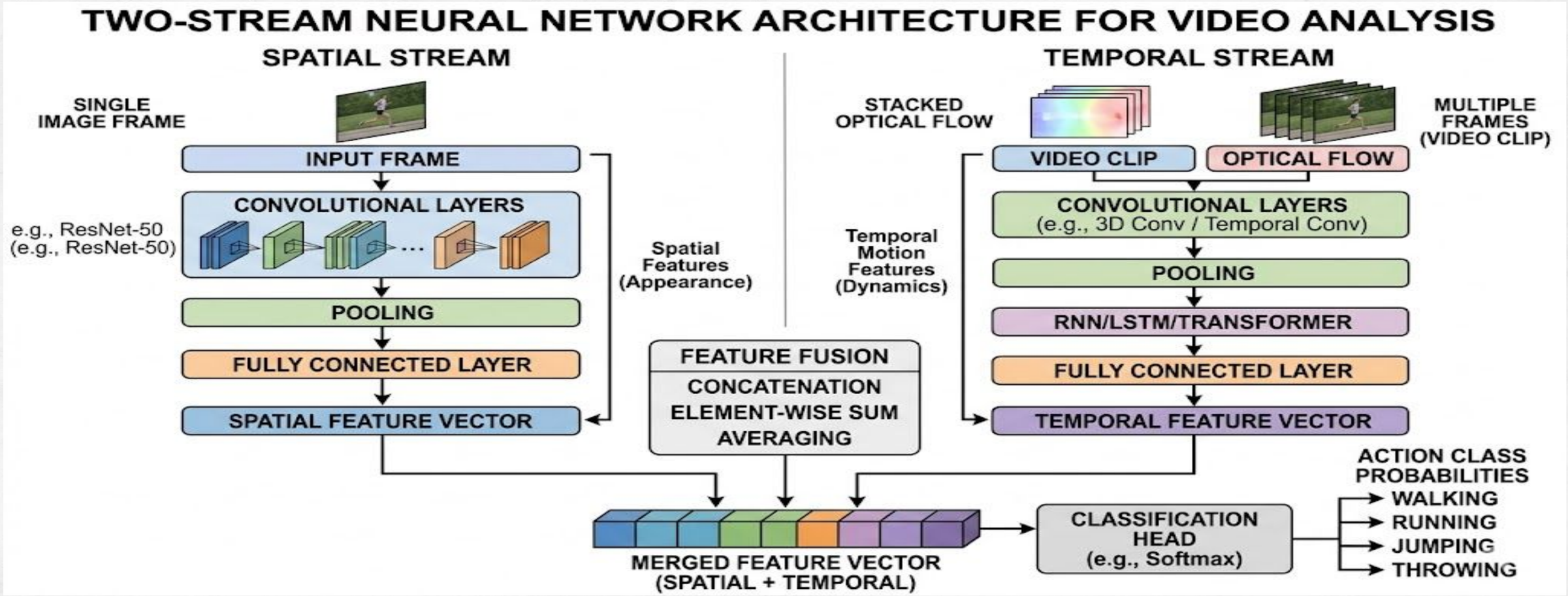
## AI-Powered Threat Detection Pipeline

1. CAMERA FEED (LIVE)
Step 1
2. EXTRACT FRAMES
Step 2
3. PREPARE INPUT
Step 3
4. SPATIAL ANALYSIS
Step 4
5. TEMPORAL ANALYSIS
Step 5
6. FEATURE FUSION
Step 6
7. VISION TRANSFORMER
Step 7
8. CLASSIFICATION
Threat Person Vehicle
Step 9
9. CONFIDENCE CHECK
Step 10
10. 5s VERIFICATION
5s
11. ALERT GENERATION

# Model Architecture

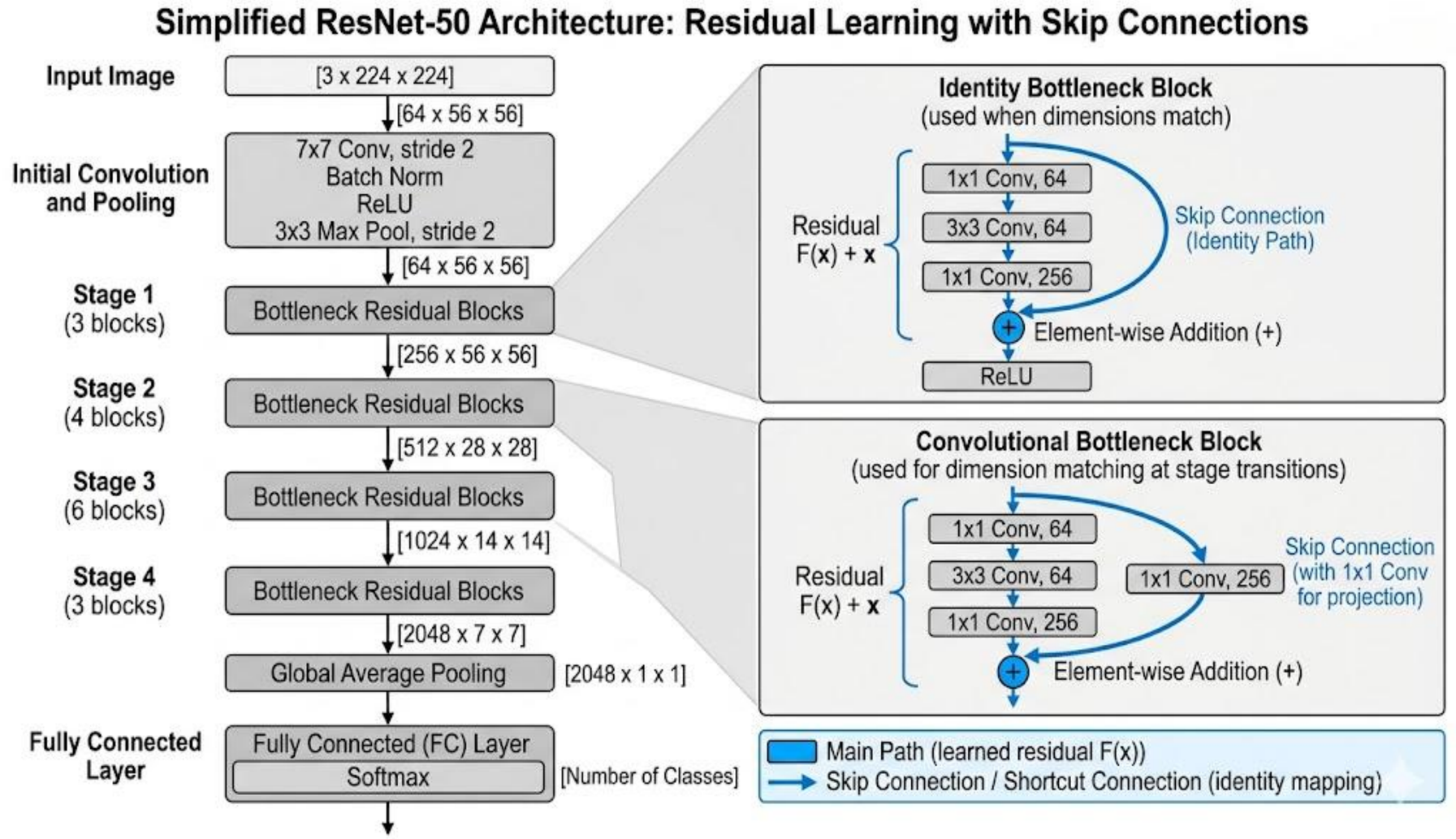


# Two-Stream Architecture



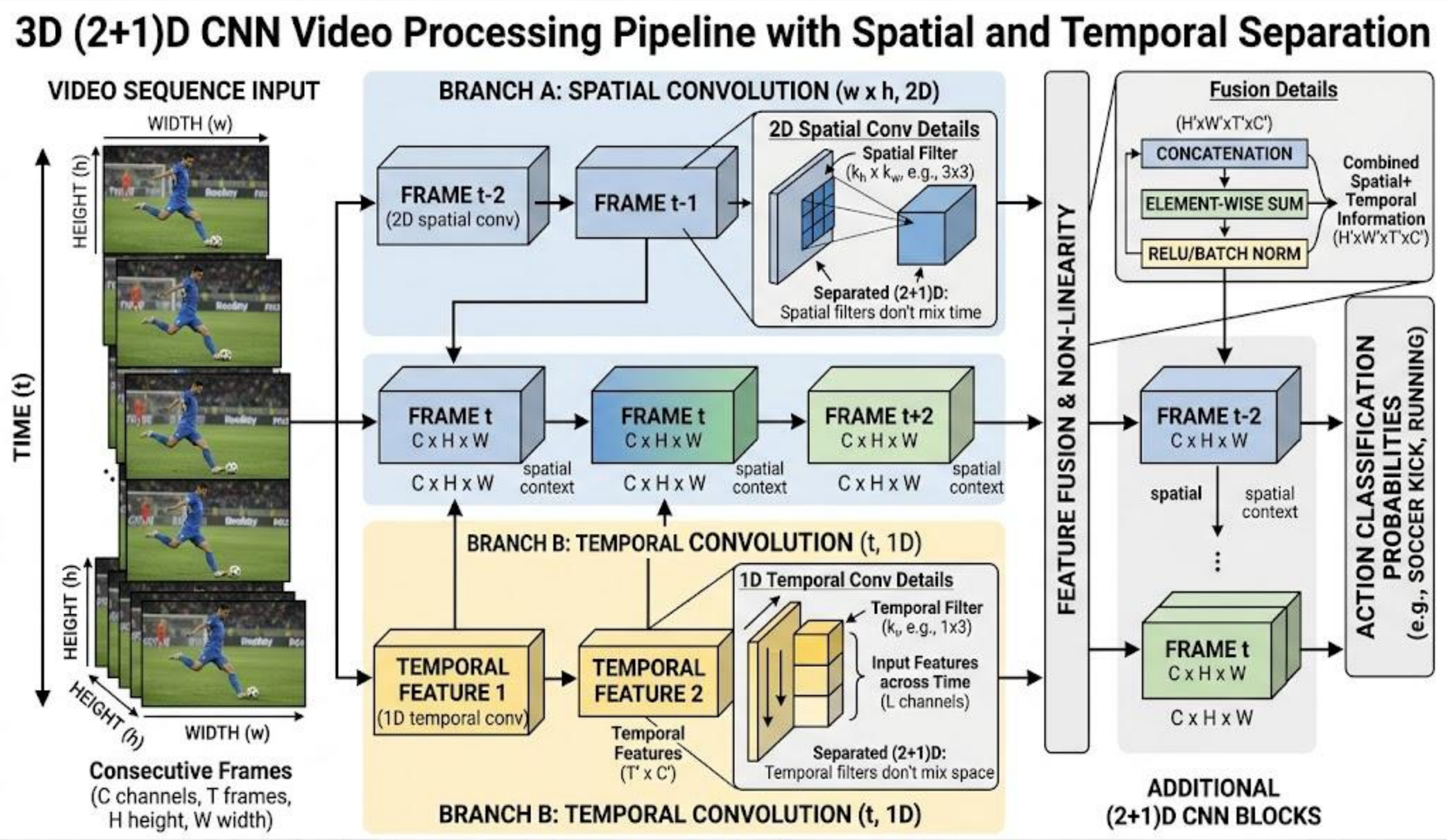

- **Formulation:**
  - ➔ Ffinal=Fuse(Fs,Ft)

# Spatial Stream (ResNet-50)



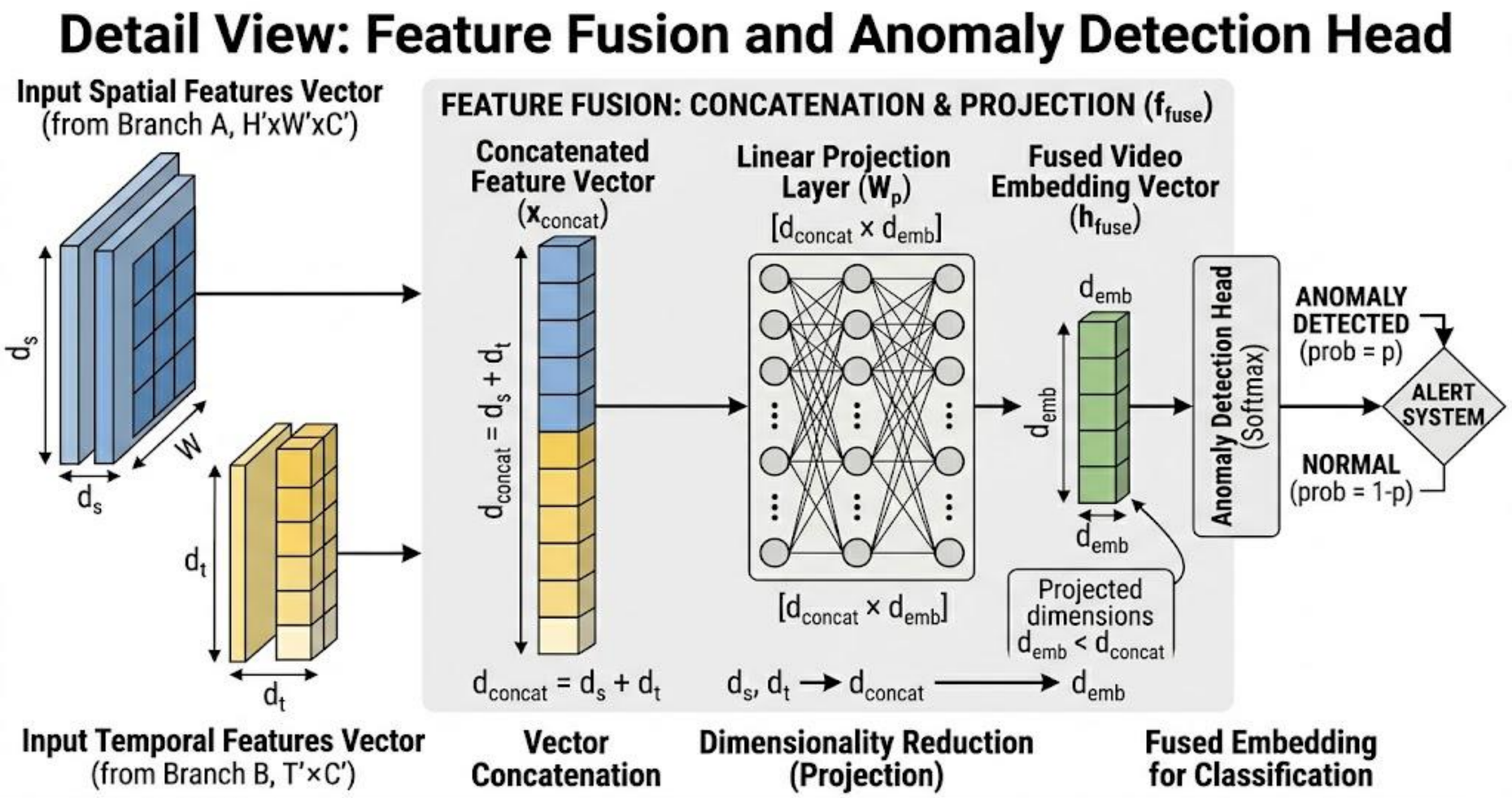

- **Spatial convolution:**
  - ➔ **Fs1 = f (Ws ∗ xt + bs)**
- **Residual Learning:**
  - ➔ **Fs = F (Fs1 + xt)**
- **Final Spatial Output:**
  - ➔ **Ys = f (Ws ∗ Xt + bs) + Xt**

# Temporal Stream (R(2+1)D)



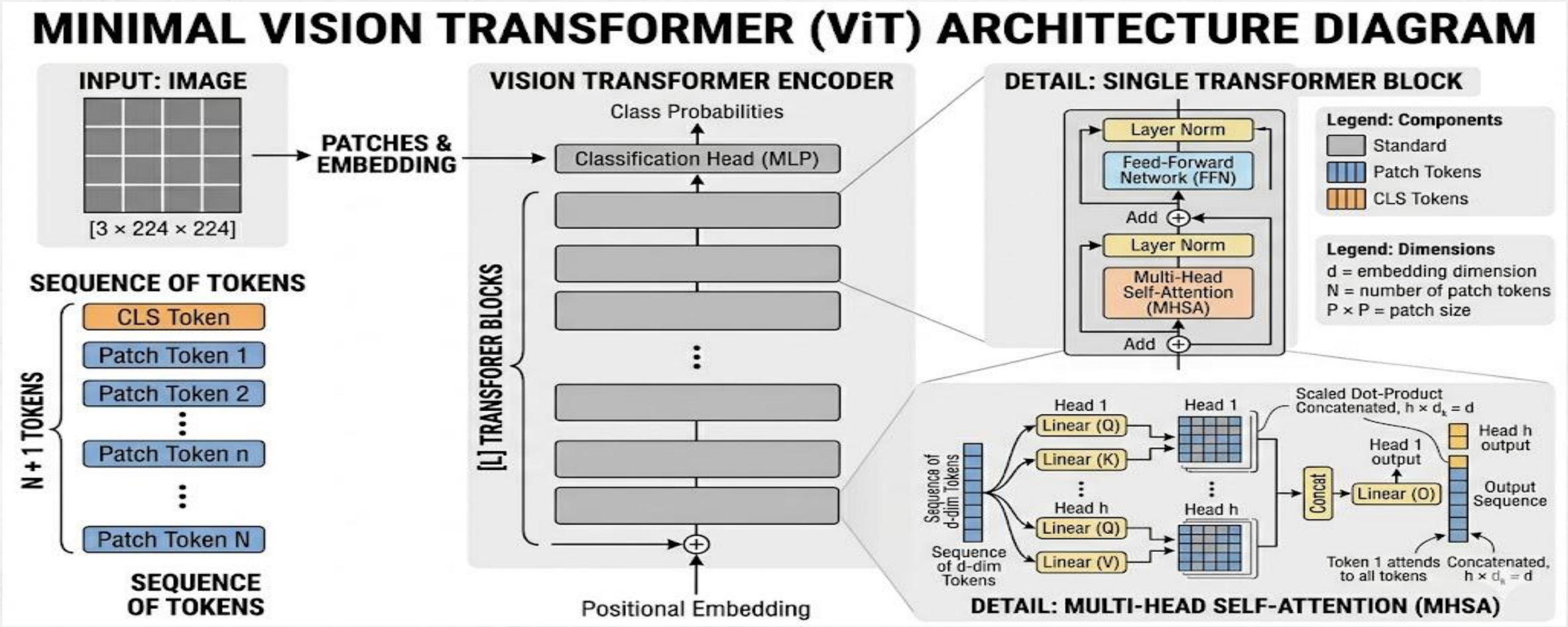

- **Spatial convolution:**
  - ➔ st= fs (Ws ∗ xt)
- **Temporal convolution:**
  - ➔ Ft = ft (Wt * Fst + bt)
- **Residual Learning:**
  - ➔ Yt = F (Fs1 + xt)
- **Final Spatial tOutput:**
  - ➔ Ys = ft (fs (Ws ∗ X)) + X

# Feature Fusion



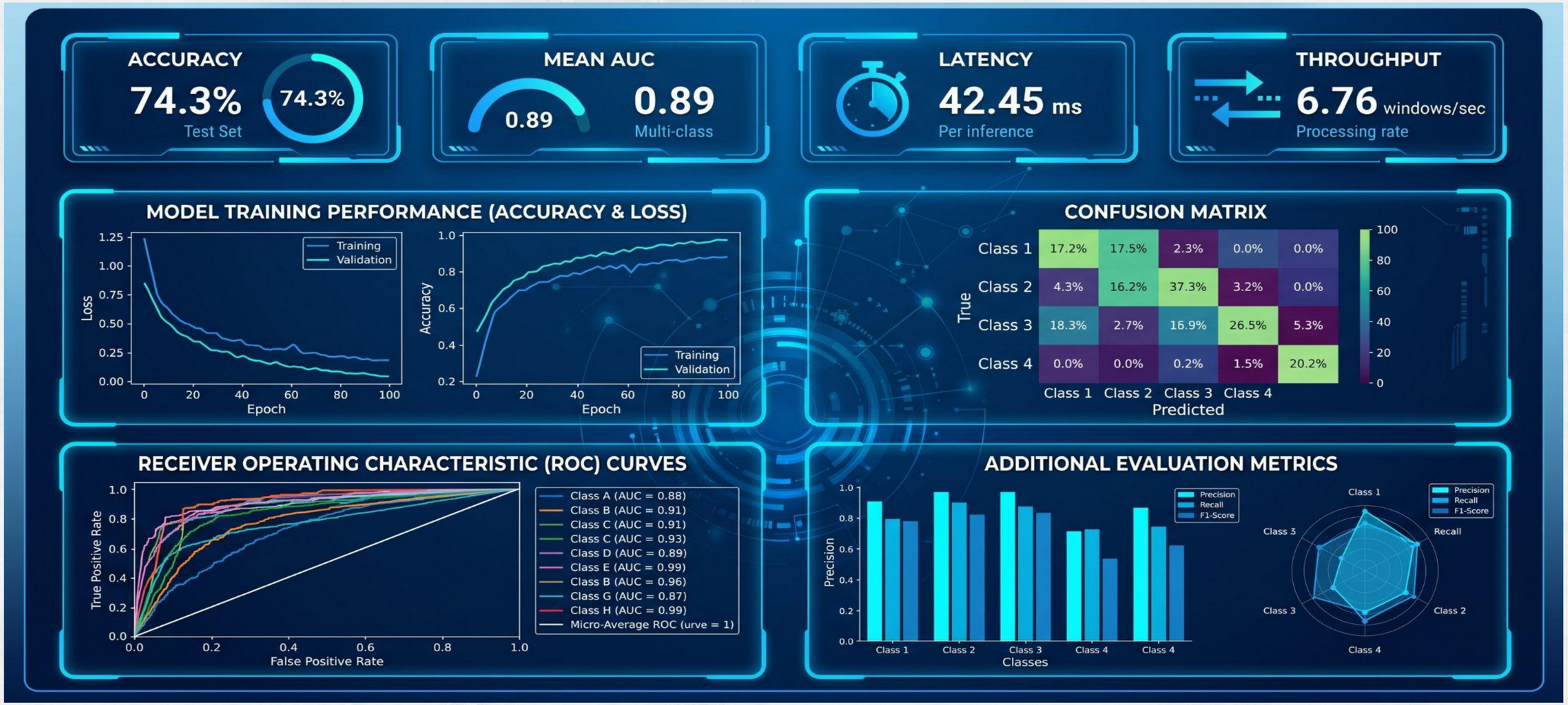

- **Equation (concatenation):**
  - ➔ F = [Fs;Ft]
- **Then projection:**
  - ➔ F′ = WF + b

# Vision Transformer (ViT)




- Attention(Q,K,V)=softmax{Q*K(Transpose) / square root (d)}*V

# Hybrid CNN-ViT Model Training process



# Multi-Scenario Anomaly Detection (MSAD) Dataset

## ↑ Pros:

- Large-scale dataset for anomaly detection.
- Contains multiple surveillance scenarios and camera angles.
- Good for detecting whether something is normal or abnormal.

## ↓ Cons:

- No detailed activity classifications for example (theft, fire, fighting).
- Limited for training models that need specific alerts per activity type.

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# UCF Crime Dataset (Chosen Dataset)

## Pros:

- Contains large-scale real-world CCTV footage.
- Multiple activity classes: theft, fighting, fire, assault, etc.
- Suitable for training models for specific alerts.
- Well-documented and widely used in academic research.

## Cons:

- Requires preprocessing to remove corrupted videos and balance classes.
- Requires feature extraction due to use of raw videos.

## Why We Chose it:

- Real CCTV footage.
- Multiple activity types for precise detection.
- Large dataset for robust model training.



# Chosen Dataset Description

- We are going to use the UCF Crime dataset









# Dataset

❑ Dataset Images number by class **Training data**

| Class | Number of images |
|---|---|
| Abuse | 19.1k |
| Arrest | 26.4k |
| Arson | 24.4k |
| Assault | 10.4k |
| Burglary | 39.5k |
| Explosion | 18.8k |
| ShopLifting | 24.8k |
| Fighting | 24.7 |
| Normal | 948k |
| RoadAccidents | 23.5k |
| Robbery | 41.5k |
| Shooting | 7.1k |
| Stealing | 44.8k |
| Vandalism | 13.6k |

- Total Training images in dataset **1,266,345 images/frame**

# Dataset

❑ UCF Dataset Images number by class **Test data**

| Class | Number of images |
|---|---|
| Abuse | 297 |
| Arrest | 3365 |
| Arson | 2793 |
| Assault | 2657 |
| Burglary | 7657 |
| Explosion | 6510 |
| ShopLifting | 7623 |
| Fighting | 1231 |
| Normal | 65k |
| RoadAccidents | 2663 |
| Robbery | 835 |
| Shooting | 7630 |
| Stealing | 1984 |
| Vandalism | 1111 |

- Total Training images in dataset 111,308 **images**

Th
eft Burglary, Robbery,
Shoplifting, Stealing

# Data After Merging

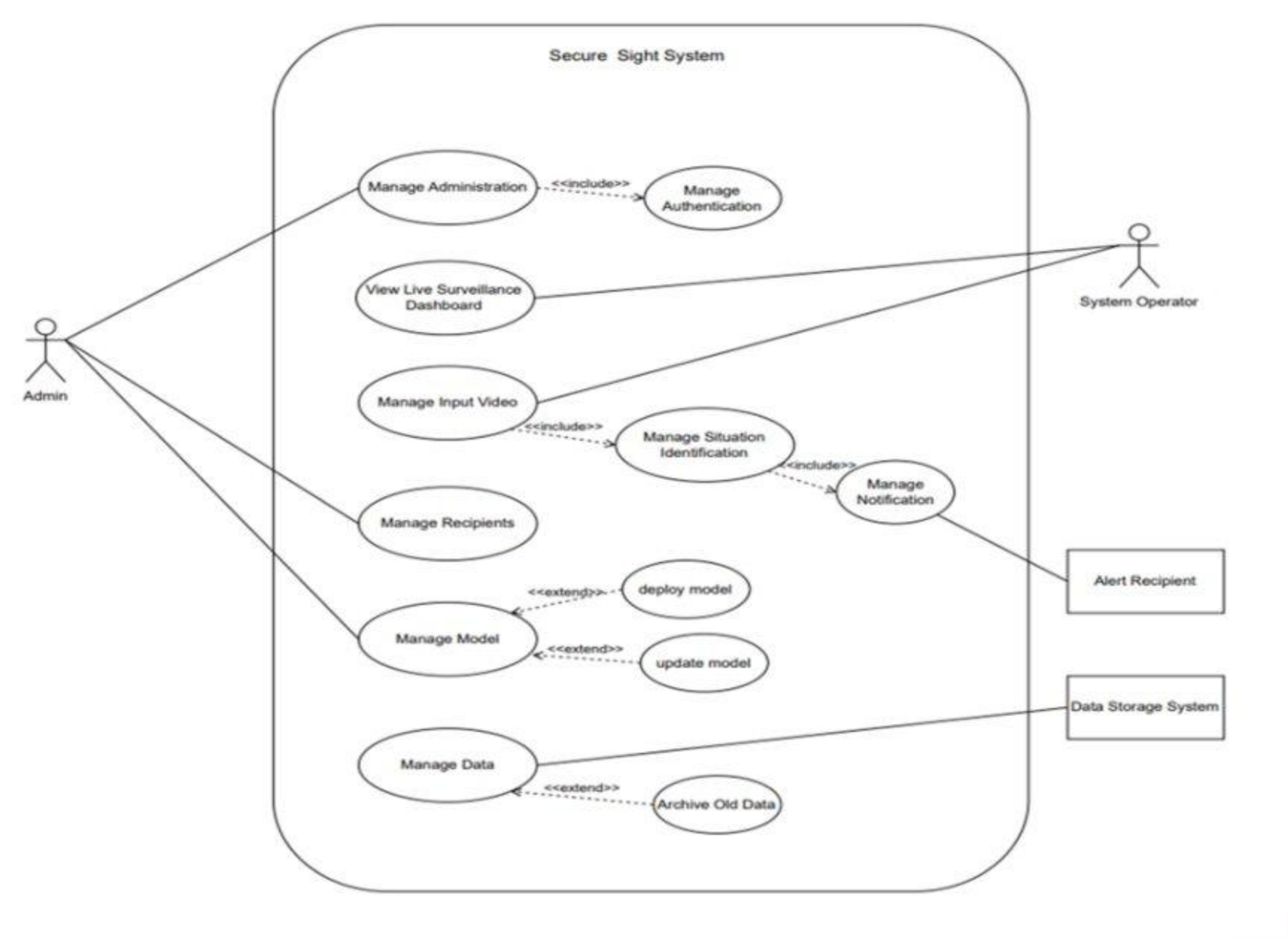
**Merged Classes**

- Theft:Burgalry,Robbery,Shoplifting,Stealing
- Violance:Abuse,Assault,Fighting
- Normal class downsampled by 40%

| Class | Number of images |
|---|---|
| Theft | 150.6k |
| Violence | 54.2k |
| Arrest | 26.4k |
| Arson | 24.4k |
| Explosion | 18.8k |
| RoadAccidents | 23.5k |
| Shooting | 7.1k |
| Vandalism | 13.6k |
| Normal | Downsampled* |

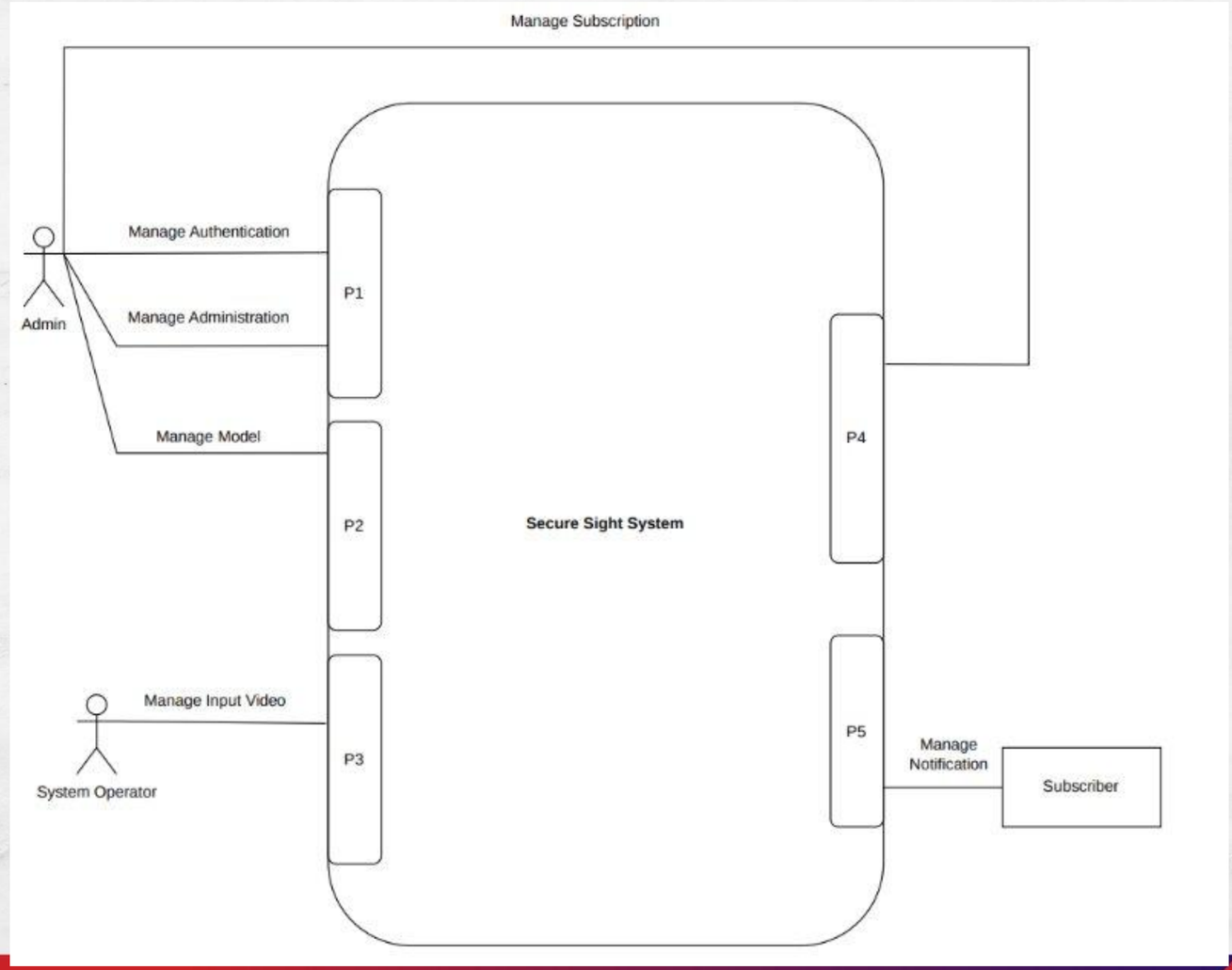
# UML Diagrams

- Use Case
- Diagram



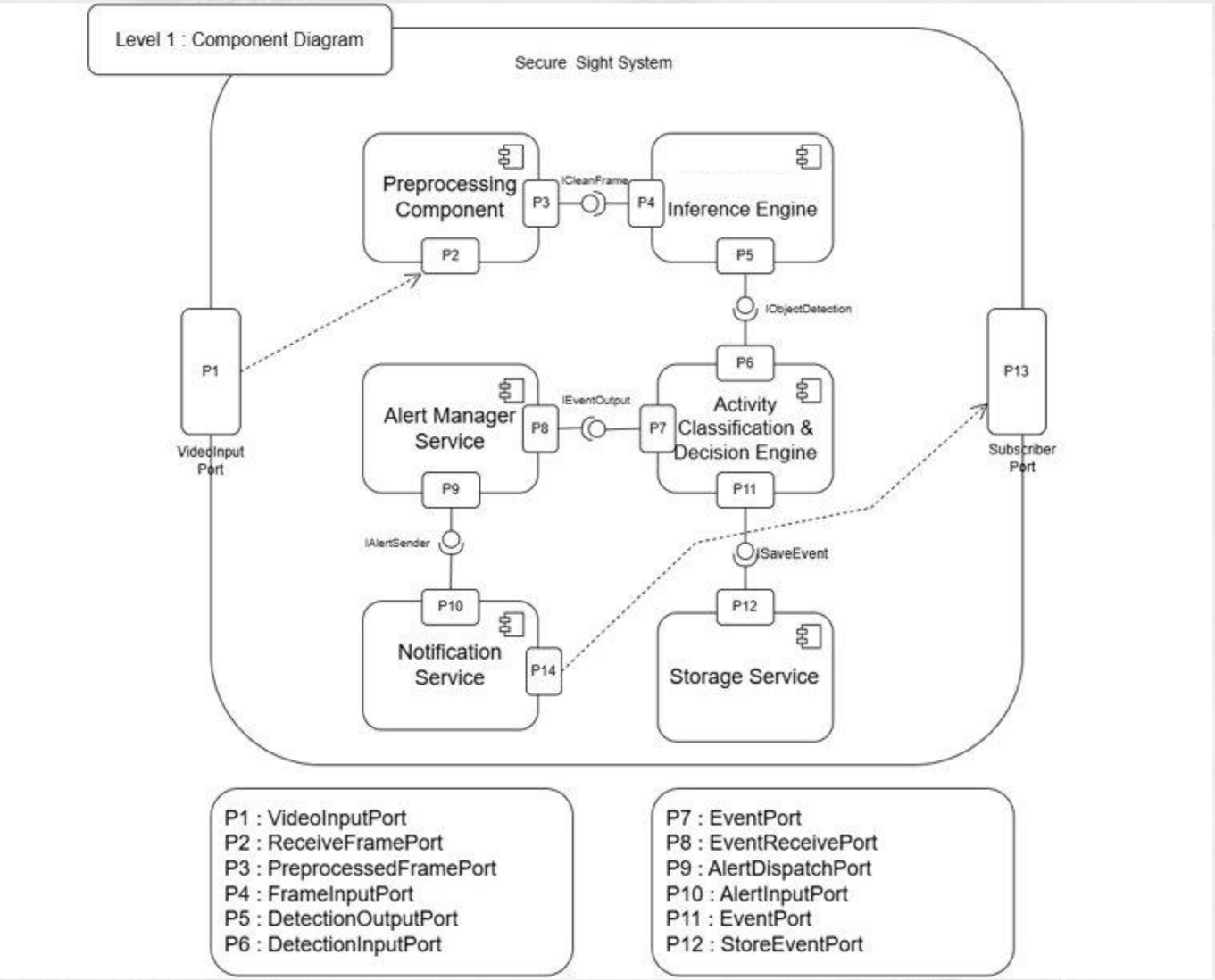
# UML Diagrams

- Level 0 : Context
- Diagram

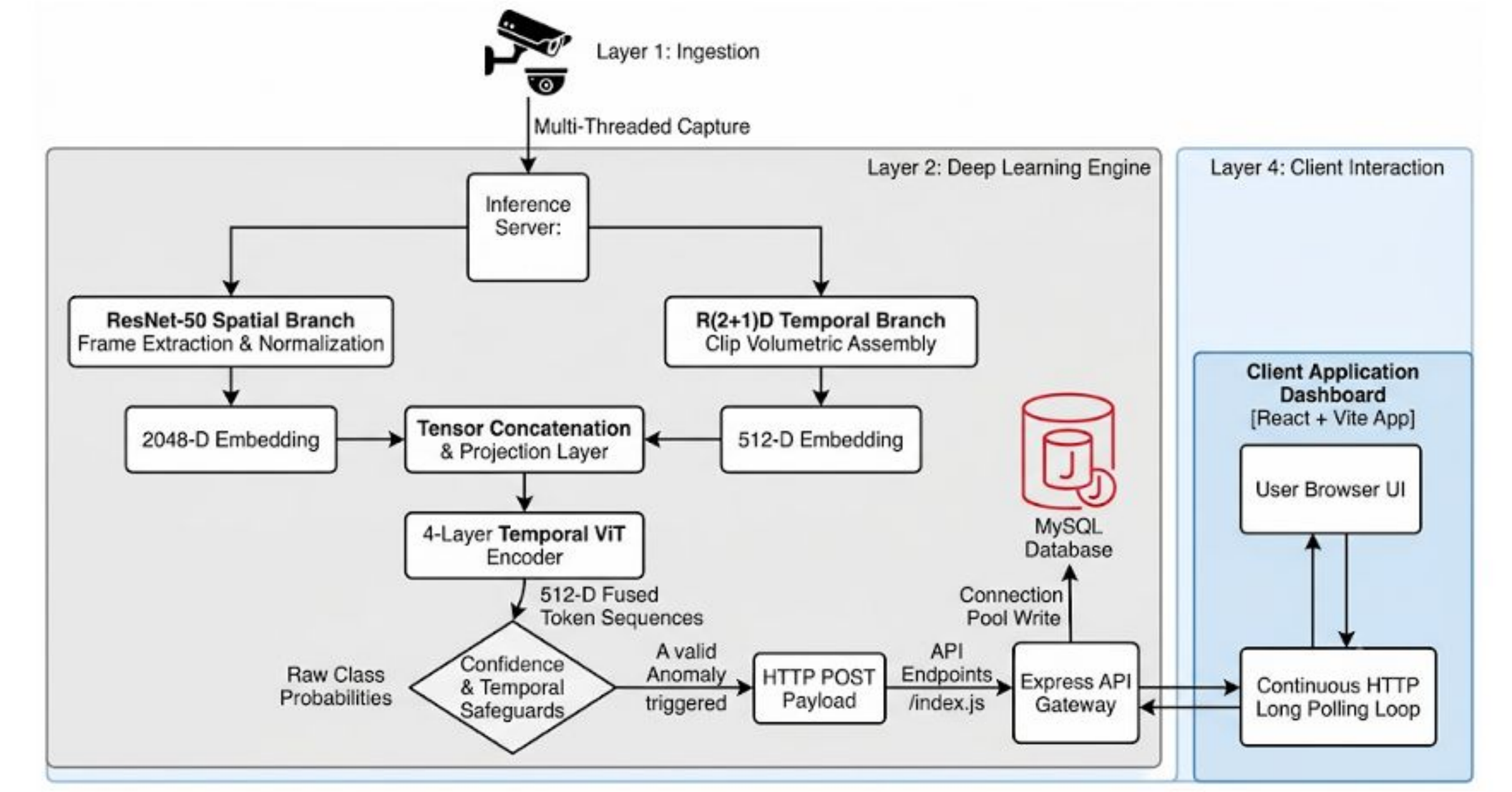


# UML Diagrams

- Level 1 : Component
- Diagram



# Implementation



# Implementation

## Web System Architecture

LIVE MODE

Integrated Data Flow & Backend API Gateway

### Data Ingestion

- RTSP Stream Capture
- Multi-threaded Pipeline
- Frame Downsampling

SOURCE
**IP Cameras (Edge)**

### Backend API Gateway

| Routing Logic | Protocol |
|---|---|
| **Async Dispatcher** | **HTTP/WS** |
| **Auth** | **Load** |
| **JWT Tokens** | **Balanced** |

Manages request queuing and response caching

### System Integration

- Alert Dispatch Hub
- Service Orchestration
- Pub/Sub Messaging

TARGET
**Responders / DB**

# Implementation

CHAPTER 5: IMPLEMENTATION DETAILS

## AI Pipeline & Optimization

LATENCY 42ms

FRAMEWORK PyTorch

### Dual-Branch Extraction

**Spatial Branch**

Static Appearance Features

`ResNet-50 (Pre-trained)`

Output: 2048-d Vector

**Temporal Branch**

Motion Dynamics (16-Frame Clips)

`R(2+1)D Factorization`

Output: 512-d Vector

CONCATENATION FUSION

**2560-d Concat Vector**

Linear Projection to 512-d

**Vision Transformer**

Sequence Modeling Layer

`4-Block Encoder`

### Training Strategy

- **Clip-Level Supervision**
  Precise temporal annotations over weak labels.
- **Class Weighting**
  Inverse frequency sampling to mitigate normality bias.
- **PyTorch Lightning**
  Structured training loops & hardware scaling.

PERFORMANCE

**Top-1 Acc: 74.3%**

**AUC: 0.89**

# Implementation

CHAPTER 5: IMPLEMENTATION DETAILS

## Inference-Time Safeguards

DECISION ENGINE

Dual-layer verification mechanism to suppress false alarms while maintaining high detection accuracy.

### Confidence Filter

THRESHOLD LEVEL

**60%**

Predictions below **60%** confidence are automatically discarded as background noise.

### Temporal Validation

VERIFICATION WINDOW

**5 sec**

- Ensures anomaly persists consistently across consecutive frames.
- Filters short-term visual noise or momentary irregularities.
- Prevents false alerts triggered by sudden, non-threatening movements.

Input Score → Gate 1: Confidence → Gate 2: Duration → EMERGENCY ALERT

# Implementation

## Deployment & Robustness

Production Ready

### Fault Tolerance

Designed to handle network failures gracefully with localized edge intelligence.

MECHANISM

**Offline Buffering**

Queues critical alerts locally during disconnects. **Zero data loss** upon network restoration.

### Data Privacy Framework

PROCESSING MODE

**Ephemeral**

Volatile memory only. No disk persistence of raw streams.

RETENTION

**Conditional**

Clips stored **only if** anomaly alert is triggered.

### Deployment

- Containerized Services
- GPU-Accelerated Inference
- Edge / Cloud Hybrid

### Operator Dashboard

5.11

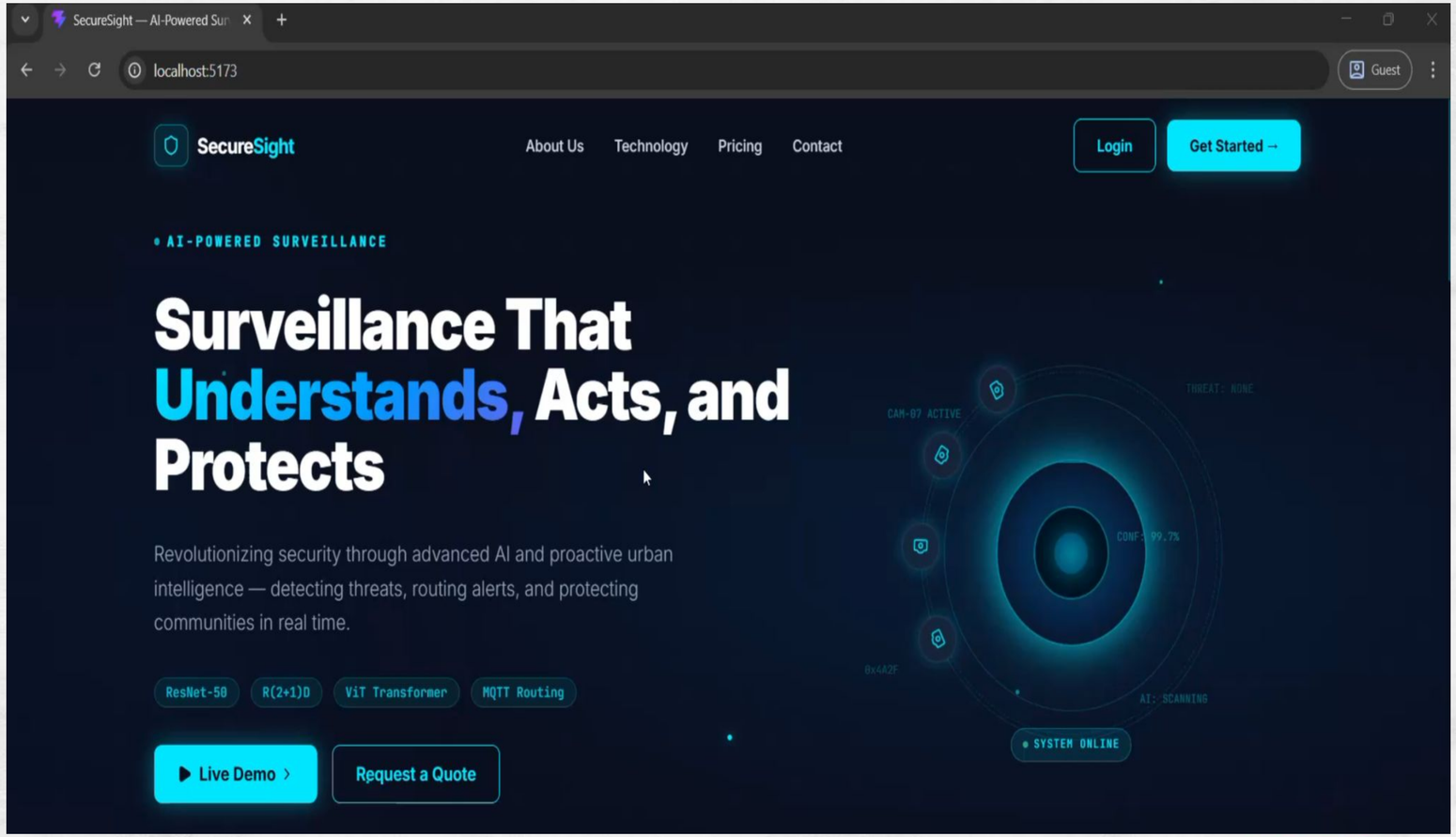
Live Feed Monitor

Real-time Stream Analysis

- Role-Differentiated Alerts
- Event Logging & History
- Threshold Configuration

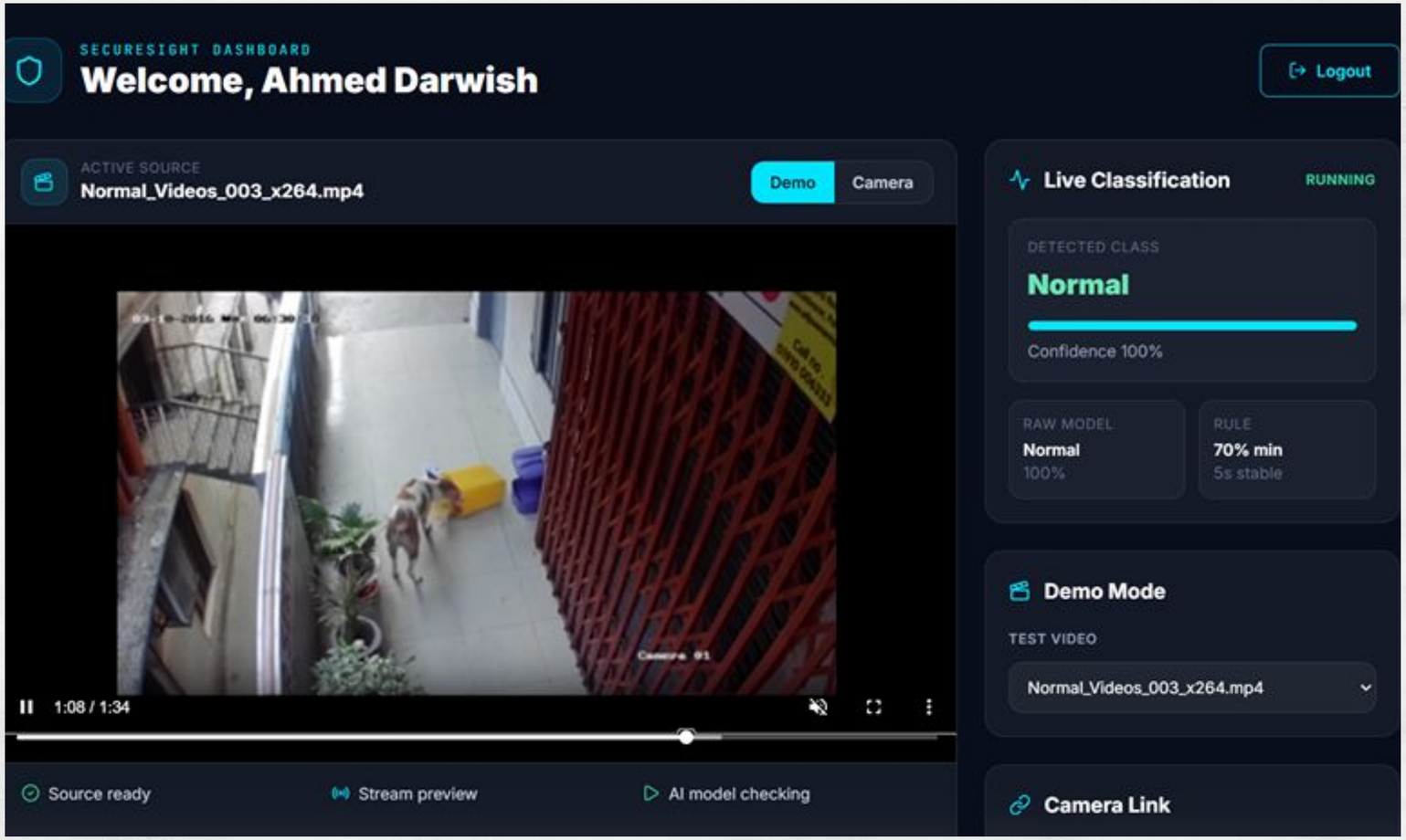
# Web Interface

- End-to-end working system
- Real-time video analysis
- Multi-class anomaly detection
- Live alert visualization



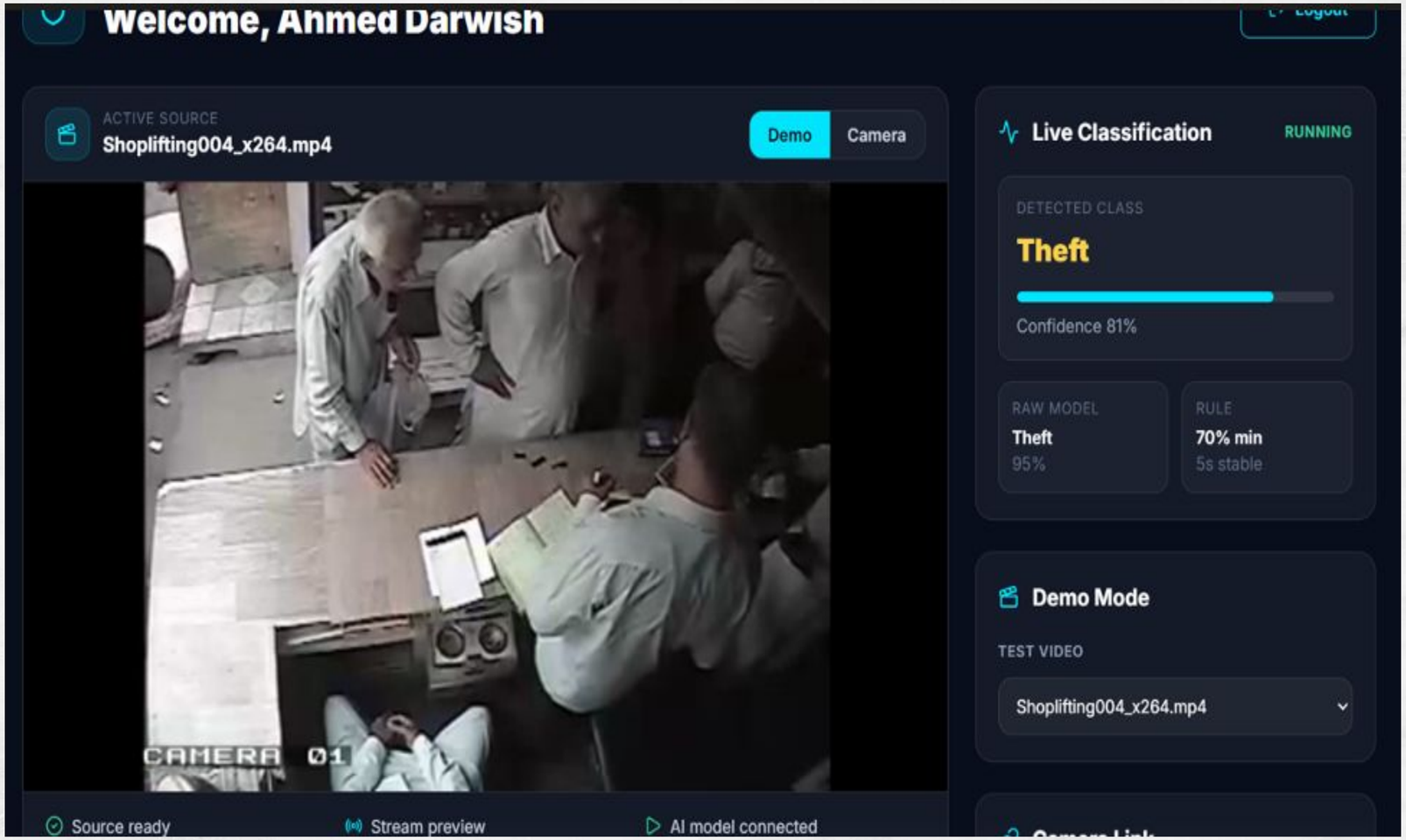
# Web Interface

- Input: CCTV / video footage
- Model processes video stream
- Detects anomaly in real-time
- Displays confidence score
- Triggers alerts



# Web Interface

- Input: CCTV / video footage
- Model processes video stream
- Detects anomaly in real-time
- Displays confidence score
- Triggers alerts



# Results

**74.3%**
**Top-1 Accuracy**
**UCF-Crime 14-class**

**0.89**
**Mean AUC**
**All anomaly classes**

**42ms**
**Processing latency**
**per window**

**Paper**
**Submitted**
**to journal**

Dataset: UCF-Crime | 128h surveillance footage | 14 classes | 70/15/15 stratified split

**Written Paper**
**Two-stream + ViT framework**
**paper accepted for journal submission**

# Comparison with Prior Work & Optimizer Study

**TABLE 3 — Comparison vs. Prior Methods**

| Method | AUC |
|---|---|
| Sultani et al. 2018 | 75.41% |
| Zhong et al. 2019 | 82.12% |
| Tian et al. 2021 | 85.33% |
| **SecureSight (Ours)** | **89.0%** |

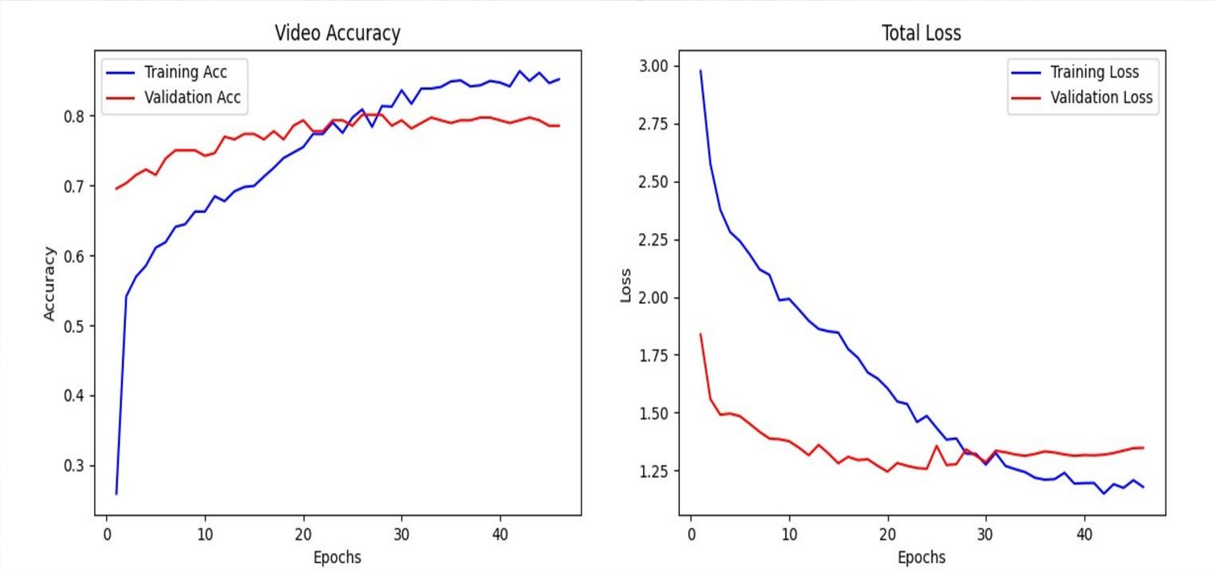
**Supervised clip-level annotations vs. weakly supervised prior work*

**TABLE 4 — Meta-Heuristic Optimizer Results**

| Optimizer | Val Acc |
|---|---|
| **Hand-tuned (baseline)** | **74.3%** |
| GWO | 70.8% |
| PSO | 70.2% |
| HHO | 69.9% |
| SMO | 69.4% |
| SFLA | 68.7% |

**All 5 optimizers failed to beat hand-tuned config**
**Negative result — reported transparently**

# Training Convergence

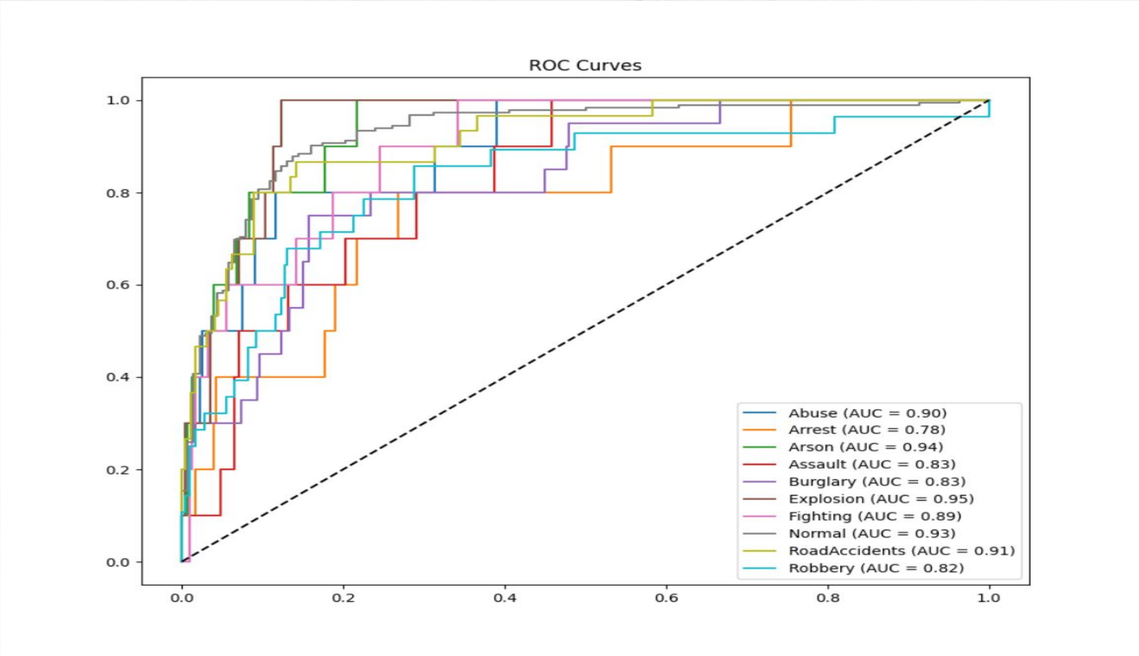



*Training & Validation Curves — 60 epochs*

**TABLE 2 — Ablation: Component Contributions**

| Model Configuration | Val Acc (%) |
|---|---|
| ViT only — no annotations | 57.2 |
| ViT only — with annotations | 70.3 |
| ResNet-50 spatial only | 63.5 |
| R(2+1)D temporal only | 67.1 |
| Two-stream fusion (no ViT) | 72.8 |
| **Two-stream + ViT (OURS)** | **74.3** |

# Per-Class AUC — ROC Analysis




*ROC curves — UCF-Crime test set*

**TABLE 1 — Per-Class AUC Values**

| Class | AUC |
|---|---|
| Abuse | 0.90 |
| Arrest | 0.78 |
| Arson | 0.94 |
| Assault | 0.93 |
| Burglary | 0.83 |
| Explosion | 0.95 |
| Fighting | 0.89 |
| Normal | 0.93 |
| RoadAccidents | 0.91 |
| Robbery | 0.82 |
| **Mean AUC** | **0.89** |

*Lowest: Arrest (0.78) & Robbery (0.82) — visual similarity to normal activity*

# paper acceptance letter

Dear Mohamed A hassona,

We are pleased to inform you that your paper A Real-Time Intelligent Surveillance System for Video Anomaly Detection Using Two-Stream Deep Learning Fusion and Vision Transformers has been accepted for online presentation at AiA26 International Conference.
Congratulations!

To proceed, we kindly ask you to:

- Submit the camera-ready version of your paper, incorporating the reviewers' comments and following the conference formatting guidelines.

- Prepare your online presentation, ensuring clear audio/video quality and adherence to the conference instructions (duration : 10 min, format : MP4, email to : aia-doctoralday@uae.ac.ma).

- It is mandatory to be present online during your session, especially for the live Q&A discussion following your presentation, to interact with the audience and answer questions.

Please ensure that the camera-ready version is submitted by April 1st, 2026.

Additional information regarding the presentation schedule and technical guidelines will be shared soon.

We look forward to your participation and your online presentation.

Kind regards,
AiA26 Steering Committee.

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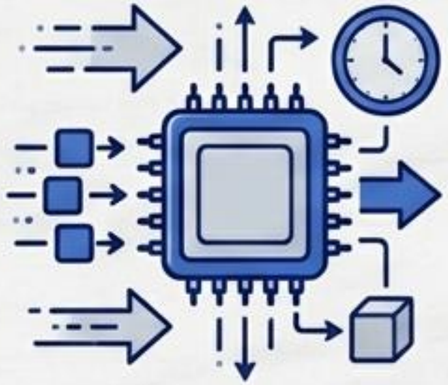
Microsoft Corporation
One Microsoft Way
Redmond, WA 98052

# Challanges




## Data Quality

- Accuracy & Completeness
- Consistency
- Validation
- Standardization



## Real-Time Processing

- Low-Latency Ingest
- In-Stream Analytics
- Instant Decisioning

## False Positives/Negatives

- Minimize Incorrect Alerts
- Reduce Missed Threats
- Optimize Thresholds







## System Integration

- Seamless API Connections
- Cross-Platform Sync
- Data Silo Elimination

# Solutions

## DATA QUALITY SOLUTIONS




PREPROCESSING PIPELINE

## REAL-TIME PROCESSING SOLUTIONS




Latency Drop: 150ms → 28ms

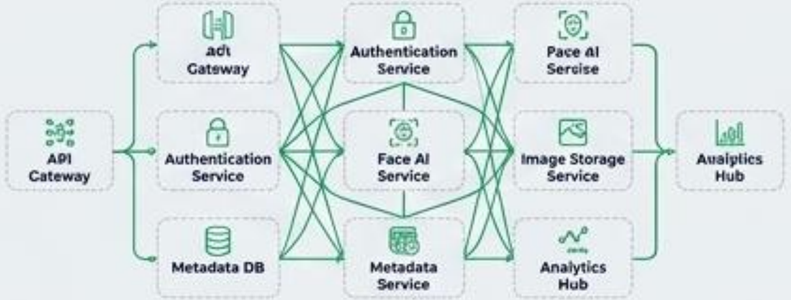
PERFORMANCE OPTIMIZATION

## FALSE ALERTS REDUCTION




THRESHOLD STRATEGY

## SYSTEM DEPLOYMENT SOLUTIONS




MICROSERVICES ARCHITECTURE

# Project Timeline

**Phase 1:** Project Initiation (Research & Scope)

**Phase 2:** Planning & Design (Strategy & Outlines)

**Phase 3:** Execution (Development & Implementation)

**Phase 4:** Integration (System Assembly & Testing)

**Phase 5:** Final Review (Validation & Quality Control)

**Phase 6:** Launch (Deployment & Delivery)

# Team Contributions

## MOHAMED AHMED
**Software Engineer & Team Lead**

Coordinated overall architecture, managed system integration.

## OMNIA ADEL
**AI Engineer**

Co-developed spatial apearance pipelines, optimized pre-trained ResNet-50 backbone.

## TASNEEM IBRAHIM
**AI Engineer**

Assisted with temporal data training, structured dataset preprocessing pipelines.

## ABDULLAH ABDALGAWAD
**AI Engineering Lead**

Designed feature extraction architecture, engineered R(2+1)D and ResNet-50 temporal pipelines, led deep learning model training.

## KHALID ABDELSALAM
**Software Engineer**

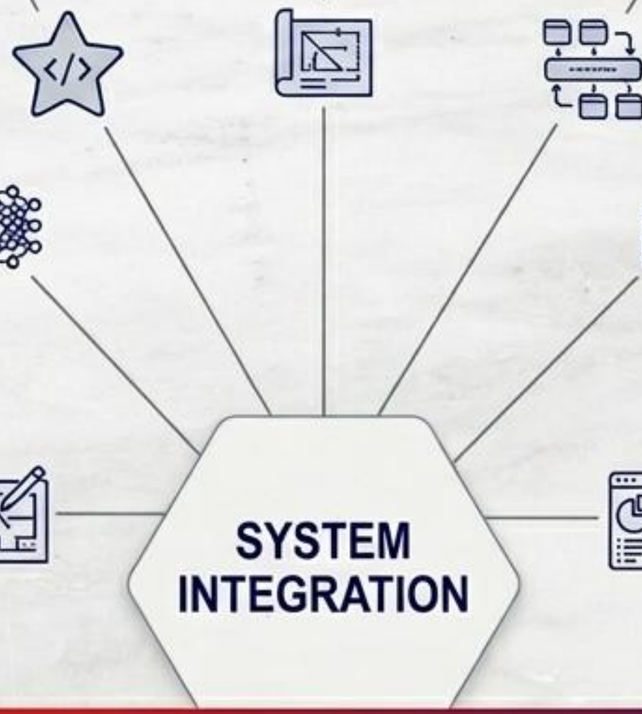
Implemented local data optimized management rules, offline edge buffering mechanisms.

## AHMED MOSTAFA
**Software Engineer**

Co-developed application, englneered system data pathways, designed system architecture.

## MARIAM ELRAFEI
**Software Engineer**

Designed live operator dashboards, handled system latency profiling.



# Limitations



## DETECTION ACCURACY

Depends on camera quality & environmental conditions



## CROWD PERFORMANCE

May decrease in highly crowded scenes




## HARDWARE RESOURCES

Real-time processing requires adequate resources




## DATA REQUIREMENTS

Some rare object classes may require additional training data

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# Future Work

**Support for Multiple Camera Streams**

Multi-stream support: Up to 50 inputs

**Cloud-Based Deployment**

Scalable Cloud Architecture

**Mobile Application Integration**

Native iOS & Android apps

**Advanced Predictive Analytics**

Pre-emptive Threat Scoring

**Continuous Learning & Model Updates**

Automated On-Device Training

**Enhanced Threat Detection Capabilities**

Multi-modal sound & motion logic

# Conclusion




Computer Vision + Deep Learning = Smarter, Reliable Surveillance

# References


- UCF-Crime Dataset – Sultani et al., CVPR 2018
- UCA Annotation Dataset – Yuan et al., arXiv 2023
- ResNet-50 – He et al., CVPR 2016
- I3D – Carreira & Zisserman, CVPR 2017
- R(2+1)D – Tran et al., CVPR 2018
- Vision Transformer (ViT) – Dosovitskiy et al., arXiv 2020
- Graph Convolutional Label Noise Cleaner – Zhong et al., CVPR 2019
- RTFM Method – Tian et al., ICCV 2021

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# Q&A